

STATEMENT OF WORK (SOW)

CALIBRATION/REPAIR SERVICES FOR NUWCDIVNPT TEST EQUIPMENT

1.0 Background

The Naval Undersea Warfare Center Division Newport (NUWCDIVNPT) is a major research and engineering facility of the Naval Sea Systems Command and is tasked with research, development, engineering, and logistic functions in support of numerous weapons systems requiring precision measurements in the test and evaluation, maintenance, and repair of subassemblies, assemblies, and total systems. The test equipment used by the various laboratories within NUWCDIVNPT must be calibrated to meet the stringent and traceable measured parameters to assure the subassemblies, assemblies, and total systems meet the performance requirements of its program specifications.

1.1 Places of Performance

All work will be performed at the contractor's facility.

1.2 Authorized Users

This contract is applicable to the Rapid Prototyping and Calibration Branch, Code 7013, of NUWCDIVNPT, Newport, RI.

1.3 Sponsors

NUWCDIVNPT

1.4 Types of Funding

Service Cost Center (SCC)

2.0 Scope

The contractor shall provide calibration and repair services required for NUWCDIVNPT test equipment. Equipment shall be calibrated and repaired to Original Equipment Manufacturer (OEM) specifications unless otherwise notified by the Government of a requirement for deviation. The calibration must provide data, uncertainty, and traceability to include the standards used for the calibration

2.1 Program or Systems Supported

Rapid Prototyping and Calibration Branch, Code 7013,

3.0 Applicable Documents

The contractor shall perform the tasking required in Section 4.0 in accordance with (IAW) the below Applicable Documents (ADs).

NUMBER	TITLE	REVISION/ DATE	APPLICABLE SOW TASK
3.1	Industry Standards		
3.1.1	International Organization for Standards (ISO) 17025, General Requirements for the Competence of Testing and Calibration Laboratories	November 2017	ALL
3.1.2	National Conference of Standards Laboratories/American National Standards Institute (NCSL/ANSI) Z540.3; Requirements for the Calibration of Measuring and Test Equipment	August 2006	ALL
3.1.3	American Society for Testing and Materials (ASTM) D3951-18; Standard Practice for Commercial Packaging	July 2018	ALL

4.0 Technical Requirements

As specific needs are identified, task orders (TOs) will be issued to initiate work. TOs will be issued on a Firm-Fixed-Price basis and will identify specific requirements within the scope of the contract.

In accordance with the ADs in Section 3.0, the contractor shall provide calibration and repair services as specified in the following tasks.

4.1 Incoming Receipt and Inspection

The contractor shall perform an incoming inspection within seven (7) days of receipt of the Government's shipping container and shall photographically document any damage prior to removing equipment from the container. The contractor shall evaluate the extent of any sustained damage to equipment during shipment from the Government to the contractor's facility and submit photographic evidence of any shipping damage to the Government.

Deliverable: CDRL A001 (Technical Report, Study/Service / Receipt Inspection Report)

- 4.1.1 The contractor shall submit a quote of the cost to repair to the Government. Upon receipt of TO modification or TO issued by the Government, the contractor shall repair the damaged equipment.

4.2 Calibration

Calibration consists of comparing the performance of the Government's test and measurement equipment (Equipment) against standards of known accuracy. Adjustments to bring Equipment within specification shall be attempted based on available procedures determined by the contractor. Equipment that cannot be adjusted to meet contractor specifications may require repair IAW SOW Task 4.3.

- 4.2.1 The contractor shall provide the following calibration services, at the level determined appropriate by the contractor, in the following order of precedence:
- Keysight Standards Lab Calibration (if available)
 - Keysight Calibration + Measurement Uncertainties + Guardbanding (equivalent to ANSI Z540.3)
 - Keysight Calibration + Measurement Uncertainties (equivalent to ISO 17025)
 - Keysight Calibration (OEM level calibration)
- 4.2.2 The contractor shall provide individual measurement results that are traced through an unbroken chain of calibrations to accepted references, such as U.S. constants or the national standards of other countries correlated with U.S. national standards as held or directed by National Institute of Standards Technology (NIST) and Department of Defense (DoD) approved sources.
- 4.2.3 The contractor shall perform preventive maintenance in conjunction with the calibration of Equipment. The contractor shall clean Equipment externally, including filters, if so equipped. This maintenance shall be performed according to procedures recommended by the manufacturer, as applicable, and/or according to accepted industry practice.
- 4.2.4 The contractor shall ensure that the existing proprietary software or firmware version and coefficients for each instrument remain intact, unless an updated issued version is authorized in writing by the Government.
- 4.2.5 Upon completion of each Equipment calibration, the contractor shall submit a Certificate of Calibration and Calibration Measurement Report electronically to the Government using Keysight's Support Portal. Calibration documents shall include the following:
- Manufacturer model and serial number
 - Environmental conditions at the date and time of calibration
 - Standards used during calibration and certification number
 - Out of Tolerance (OOT) conditions existing at time of calibration
 - Comments regarding how the OOT condition was resolved
- Deliverables: CDRL A002 (Acceptance Test Report (ATR)/Certificate of Calibration) and CDRL A003 (Calibration Certificate/Report/Calibration Measurement Report)
- 4.2.6 The contractor shall affix a calibration label indicating the calibration date, calibration due date, and calibration job reference to each calibrated piece of Equipment.
- 4.2.7 Calibration documentation and labels shall indicate calibration intervals specified by the Government. If no calibration interval is specified by the Government, documentation and labels will indicate the OEM-recommended cycle in months.

4.2.8 The contractor shall place tamper-resistant “Calibration Void” seals over cover screws and any readily available adjustments to limit undetected access to internal adjustments.

4.2.9 The contractor shall submit to the Government Calibration Measurement Reports as follows:

- A pre-data and post-data measurement report stating measured Equipment’s performance as received (before any adjustment) and if an adjustment is made, as shipped, will be provided as defined by calibration type.
- Depending on calibration level, point-by-point measurement uncertainty, Test Uncertainty Ratio (TUR) or identification of a Probability of False Acceptance (PFA) ≤ 2 percent is provided for calibration types that include uncertainties. If the calibration measurement report type requires an uncertainty and the measurement does not meet either a TUR of $\geq 4:1$ or provide a PFA ≤ 2 percent and Probability of False Reject (PFR) ≤ 15 percent, it shall be annotated on the calibration report for that measurement.
- If an OOT condition is found, a report will be provided stating the measured Equipment’s performance as shipped back to the Government. For OOT Equipment not requiring repair, the calibration measurement report will reflect the measured Equipment’s performance data after the adjustment has been made. If a repair need is identified and performed, the calibration measurement report will reflect the measured Equipment’s performance data before and after the repair, as well as any subsequent adjustments that were made.

Certificates of Calibration and Calibration Measurement Reports generated IAW selected calibration level will be retained by the contractor per its Data Retention Policy. The retention period begins upon the record creation date.

CDRL A002 (Acceptance Test Report (ATR)/Certificate of Calibration) and CDRL A003 (Calibration Certificate/Report/Calibration Measurement Report)

4.2.10 In the event that a Keysight calibration standard used to calibrate the Government’s Equipment is found to have an OOT condition with potential impact to the Government, the contractor shall notify the Government of the Equipment serial numbers and affected Equipment parameters that were calibrated with that standard or possibly issue a recall notice.

4.3 Repairs

Repairs consist of diagnosis and correction of Equipment malfunctions and failures, including recalibration to ensure the Equipment meets published specifications. The contractor shall provide all labor, genuine factory replacement parts, and materials necessary to return Government Equipment to operating condition. This may include replacing the unit entirely with a new or refurbished unit (e.g., repair by exchange or factory exchange) at the discretion of the contractor. Engineering improvement modifications may be installed at the time of repair upon Government issuance of a TO modification or TO. Procedures designed to minimize Equipment

failure and extend useful product life, e.g., cleaning, adjusting lubricating, inspecting, and testing, may also be performed at the time of repair per industry standards.

- 4.3.1 **Standard Repairs.** The contractor shall provide the Government a quote for repair services for review, unless the unit is under warranty or an existing maintenance agreement. Upon receipt of TO modification or TO, the contractor shall perform standard repair (STREP) services.

Deliverables: CDRL A002 (Acceptance Test Report (ATR)/Certificate of Calibration) and CDRL A003 (Calibration Certificate/Report/Calibration Measurement Report)

- 4.3.2 **Non-Standard Repairs.** The contractor shall provide a firm fixed price quotation for Non-Standard Repairs to the Government for review. Upon receipt of TO modification or TO, the contractor shall perform Non-Standard Repairs.

Deliverables: CDRL A002 (Acceptance Test Report (ATR)/Certificate of Calibration) and CDRL A003 (Calibration Certificate/Report/Calibration Measurement Report)

- 4.3.3 The contractor shall return Equipment that an OOT condition is found following repair to the Government.

Deliverables: CDRL A002 (Acceptance Test Report (ATR)/Certificate of Calibration) and CDRL A003 (Calibration Certificate/Report/Calibration Measurement Report)

- 4.3.4 **Expedited Service Requests.** Upon issuance of TO modification or TO, the contractor shall provide expedited calibration or repair services as specified in the individual TO modification or TO.

5.0 Progress Reports

Reserved

5.1 Contract Status Report

Reserved

5.2 Training

Reserved

6.0 Government Furnished Information

Reserved

7.0 Government Furnished Property

The Government may provide Government Furnished Property (GFP) as listed in an attachment to orders placed against this contract. The contractor shall adhere to DFARS Clause 252.245–7005 for the management and reporting of Government Property and shall use the GFP Module of the Procurement Integrated Enterprise Environment (PIEE) as specified in the clause. In addition, the Contractor shall prepare and submit a Government Property (GP) Inventory Report to capture GFP status IAW CDRL A004.

Deliverable: CDRL A004

8.0 Quality Surveillance and Performance Standards

Reserved

9.0 Information and Communication Technology (ICT) Accessibility Requirements

The Government has determined that this procurement is not Information and Communication Technology (ICT)

10.0 Security Compliance

10.1 Program Protection

SECURITY: All Contractor personnel shall adhere to the Security provisions of 32 CFR Part 117 – National Industrial Security Program Operating Manual (NISPOM). Applicable FAR, DFARS, NMCARS clauses, and NAVSEA text shall be adhered to in the performance of this contract. Security incidents shall be promptly reported through the companies Facility Security Officer (FSO), to the Technical Point of Contact (TPOC), and the Cognizant Security Office to NUWCDIVNPT Security.

Controlled Unclassified Information (CUI) including Legacy FOUO and Covered Defense Information (meeting the definition of 48 CFR 252.204–7012(a)) generated and/or provided under this contract shall be marked and safeguarded as specified in DoD Instruction 5200.48, CUI available at:

<https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/520048p.PDF>. Any product containing Covered Defense Information shall be assigned a distribution statement (distribution statements B through F) in accordance with DoDI 5230.24 (Distribution Statements on Technical Documents); and DoDI 5230.24, Enclosure 3 Procedures, available at <https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/523024p.pdf>

INFORMATION SECURITY: If the work is performed at the Contractor's facility, the Contractor shall implement and maintain security procedures and controls to prevent unauthorized disclosure of classified information and controlled unclassified information (CUI) and to control distribution of CUI in accordance with National Industrial Security Program Operating Manual (NISPOM) codifying 32 CFR Part 117, NISPOM Rule, and SECNAV M-5510.36B. If the work is performed at the Government's facility, the Contractor shall comply with facility policy.

CUI INCIDENT REPORTING AND RESPONSE: The Contractor shall promptly report any unauthorized, inadvertent, or illegal release or disclosure of CUI to the Contracting Officer's Representative / Technical Point of Contact (TPOC), Procuring Contracting Officer, and the Security Office. Contractor personnel shall coordinate this effort through the relevant industry site FSO.

PUBLIC RELEASE: Any controlled unclassified information pertaining to this contract shall not be released for public dissemination, including posting to any social media sites such as Facebook or Twitter, unless it has been approved for public release by appropriate U.S. Government authority. Proposed public releases shall be submitted for approval prior to release through the appropriate U.S. Government Office.

10.2 Operations Security (OPSEC)

OPSEC is a process that identifies critical information to determine if friendly actions can be observed by adversary intelligence systems, determines if information obtained by adversaries could be interpreted to be useful to them, and then executes selected measures that eliminate or reduce adversary exploitation of friendly critical information.

The Contractor shall develop and implement, and update and maintain an OPSEC program to protect controlled unclassified and classified activities, information, equipment, and material used or developed by the Contractor and any subcontractor during performance of the contract. The Contractor shall be responsible for the subcontractor implementation of the OPSEC requirements. The Contractor developed OPSEC program may include Information Assurance and Communications Security (COMSEC). The OPSEC program shall be in accordance with National Security Presidential Memorandum (NSPM) 28, and at a minimum shall include:

- 1) Assignment of responsibility for OPSEC direction and implementation.
- 2) Issuance of procedures and planning guidance for the use of OPSEC techniques to identify vulnerabilities and apply applicable countermeasures.
- 3) Establishment of OPSEC education and awareness training. To include initial and annual OPSEC training.
- 4) Provisions for management, annual review, and evaluation of OPSEC programs.
- 5) Flow down of OPSEC requirements to subcontractors when applicable.

While performing aboard Government sites, the contractor shall: comply with all OPSEC instructions and policies; include OPSEC as part of its ongoing security awareness program and take all required Agency training; Be responsive to the Supporting OPSEC Manager on a non-interference basis; and Protect sensitive unclassified information and activities, which could compromise classified information or operations, or degrade the planning and execution of operations performed by the Requiring Organization and contractor in support of the mission.

10.3 Electronic Spillage

Electronic Spillage (ES) is defined as a situation where information of higher classification than a system is authorized to process is introduced into that system, intentionally or otherwise. If a Contractor is determined to be responsible for an ES, all direct and indirect costs incurred by the Government for ES remediation will be charged to the Contractor.